

Unit 1: Introduction and Physical Layer

Overview of Computer Network, OSI and TCP/IP Reference Models, Guided and Unguided Transmission Media, Analog and Digital Communication, Encoding and Modulation, Nyquist Theorem, Shannon's capacity, Switching techniques- TDM, FDM.

**To transfer the data over the transmission media,
the data must be converted/transformed to
signal**

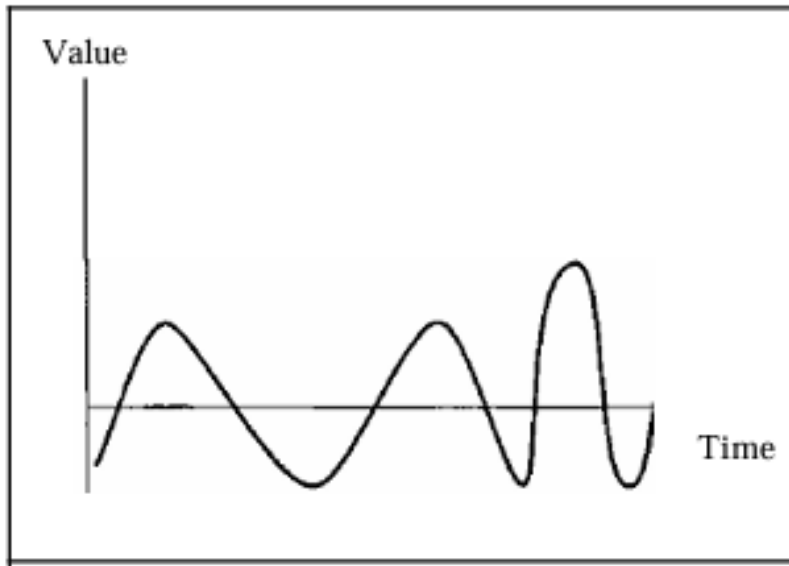
- ❖ Data can be
 - ✓ **Analog** refers to information that is continuous or
 - ✓ **Digital** refers to information that has discrete states

Example:

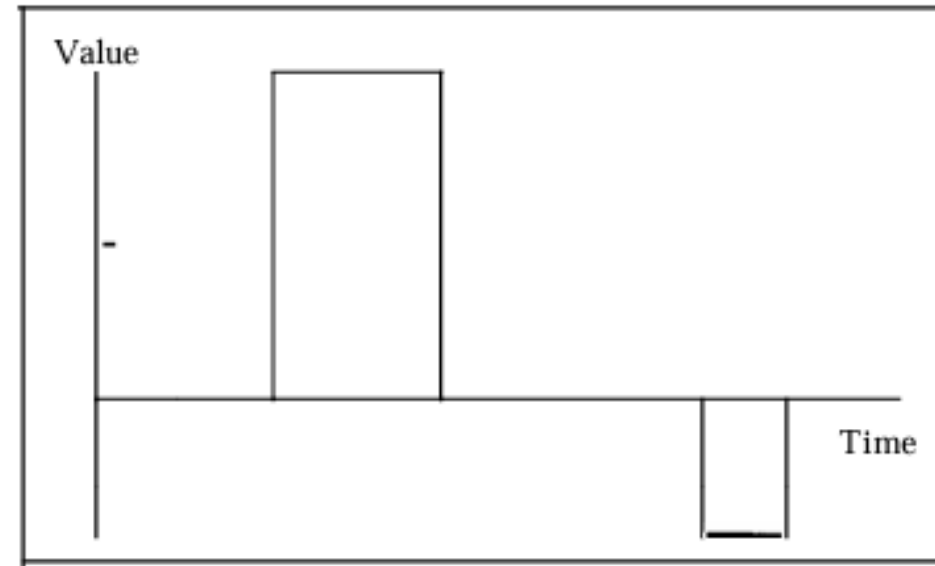
- ❖ an **analog clock** that has hour, minute, and second hands gives information in a continuous form; the movements of the hands are continuous.
- ❖ **digital clock** that reports the hours and the minutes will change suddenly from 8:05 to 8:06.

Data & Signal

Signals can be analog or digital. Analog signals can have an infinite number of values in a range; digital signals can have only a limited number of values.



a. Analog signal



b. Digital signal

Data & Signal

Conversion of Analog signal to Digital Signal



- ❖ **Sampling:** Measuring the signal at periodic time intervals
- ❖ **Quantization:** Representation of a continuous variable with the finite number of states

Classification of Signals

- ❖ periodic signal completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods. The completion of one full pattern is called a cycle.
- ❖ nonperiodic (aperiodic) signal changes without exhibiting a pattern or cycle that repeats over time.

In data communications, we commonly use periodic analog signals and nonperiodic digital signals.

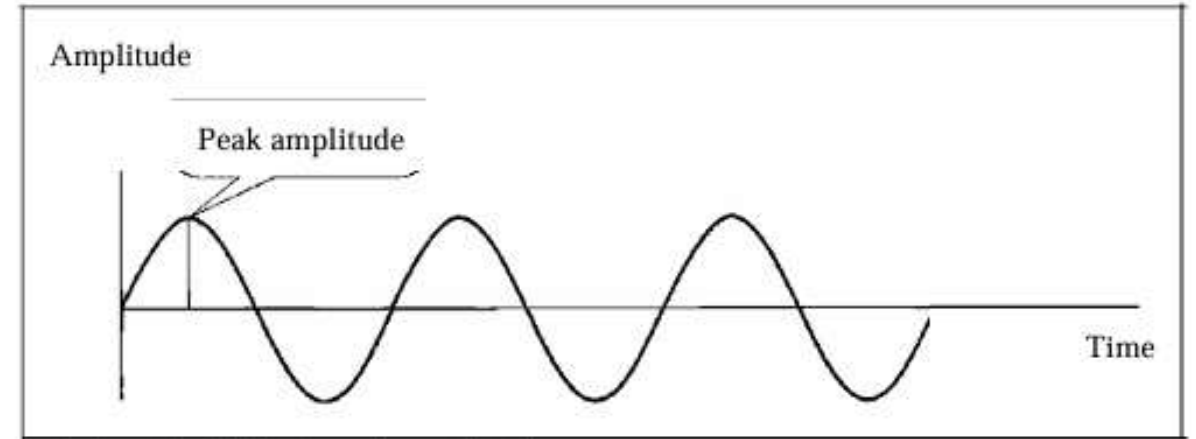
Data & Signal

All Signals have three Components

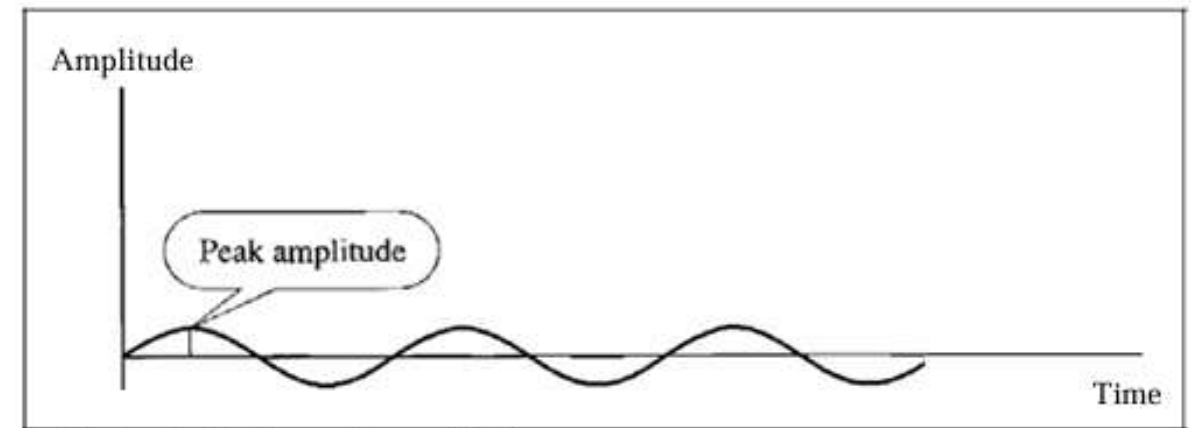
- ❖ Amplitude
- ❖ Frequency
- ❖ Phase

Amplitude

- ❖ The amplitude of a signal is the height of the wave above or below a given reference point. The amplitude is normally measured in volts.



a. A signal with high peak amplitude



b. A signal with low peak amplitude

Period and Frequency

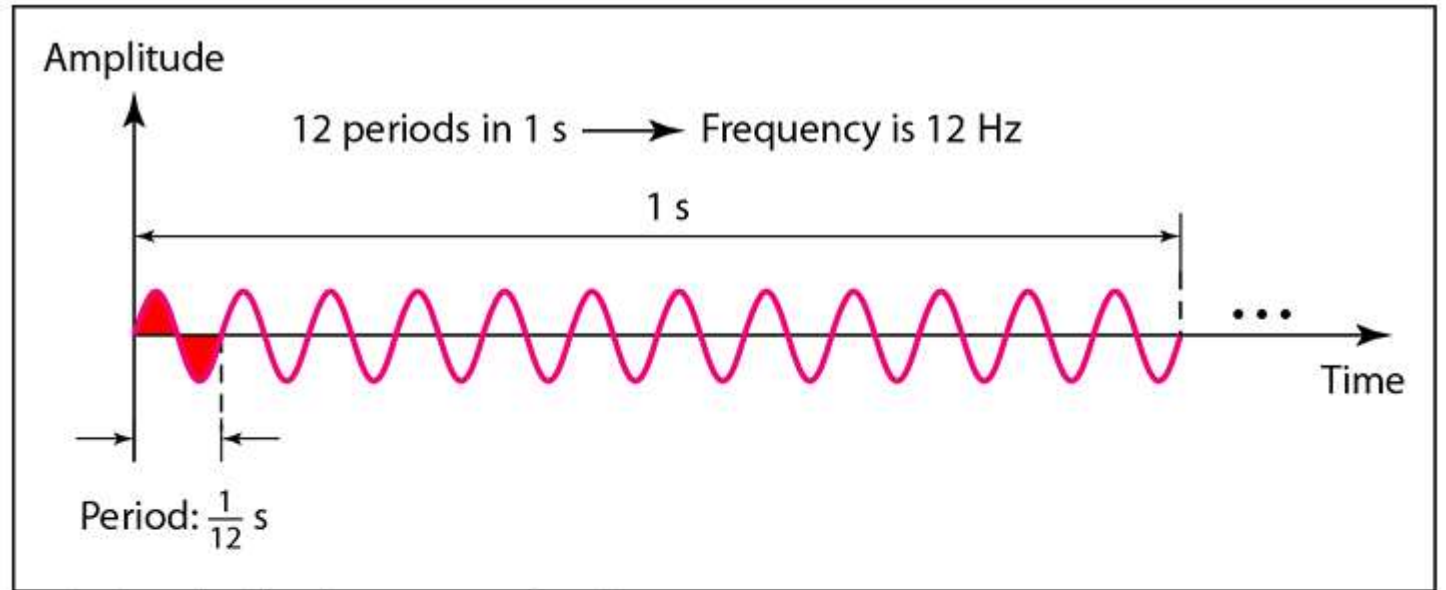
- ❖ Period refers to the amount of time, in seconds, a signal needs to complete 1 cycle. Period is formally expressed in seconds.
- ❖ Frequency refers to the number of periods in 1 s. Frequency is formally expressed in Hertz (Hz), which is cycle per second.

Frequency and period are the inverse of each other.

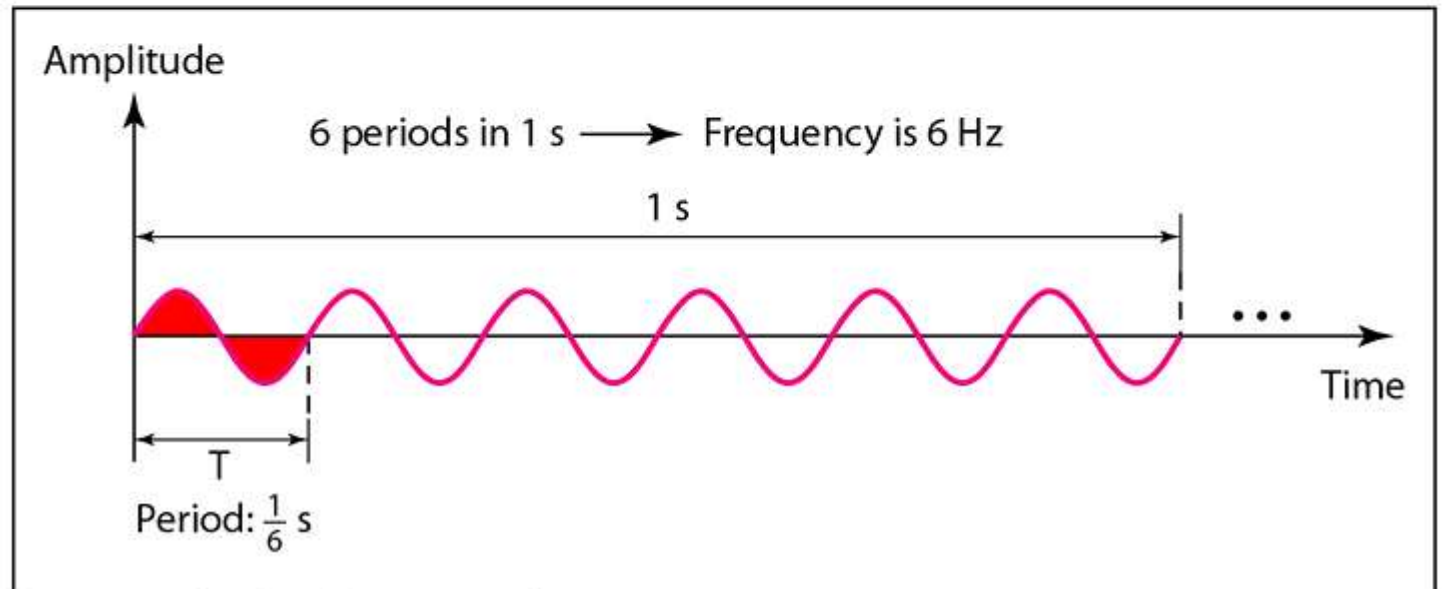
$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Data & Signal

Period and Frequency



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Period and Frequency

Units of period and frequency

<i>Unit</i>	<i>Equivalent</i>	<i>Unit</i>	<i>Equivalent</i>
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microseconds (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Data & Signal

Period and Frequency

*The power we use at home has a frequency of **60 Hz**. The period of this sine wave can be determined as follows:*

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$

The period of a signal is 100 ms. What is its frequency in kilohertz?

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$
$$f = \frac{1}{T} = \frac{1}{10^{-1}} \text{ Hz} = 10 \text{ Hz} = 10 \times 10^{-3} \text{ kHz} = 10^{-2} \text{ kHz}$$



Data & Signal

Period and Frequency

Frequency is the rate of change with respect to time.

Change in a short span of time means high frequency.

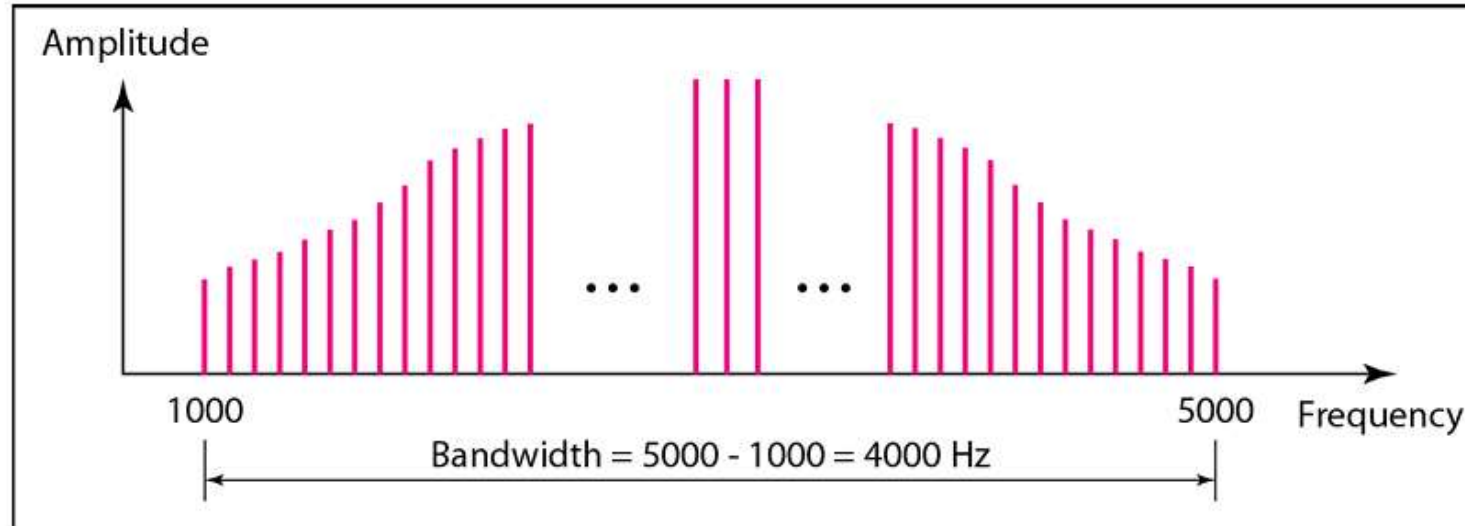
Change over a long span of time means low frequency.

If a signal does not change at all, its frequency is zero.

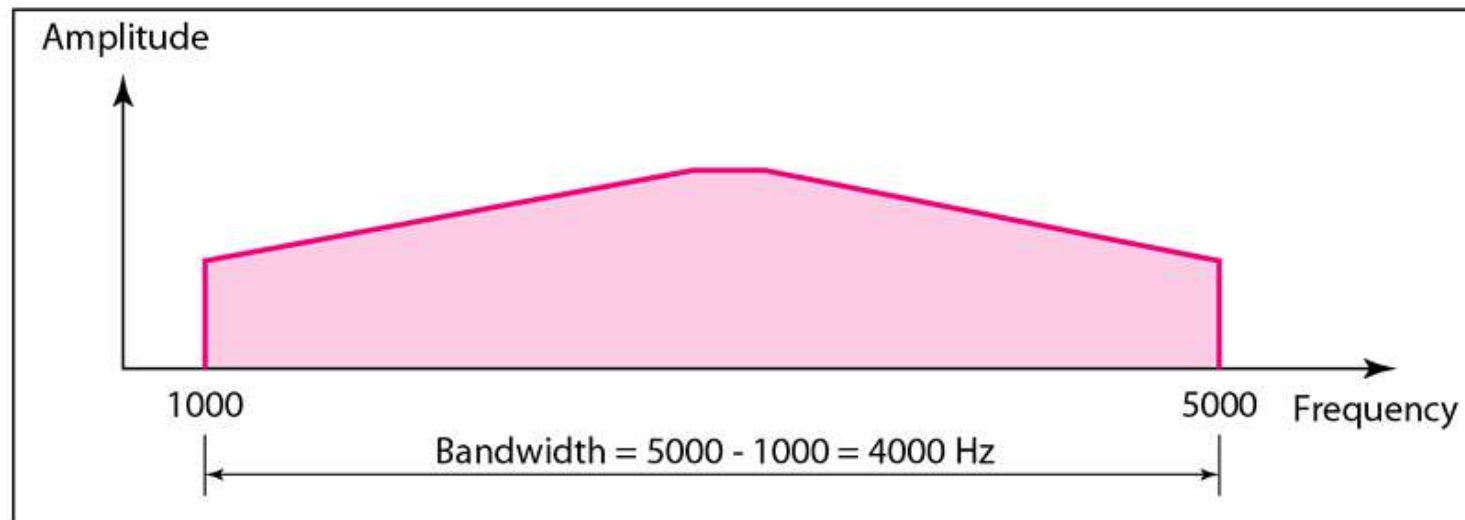
If a signal changes instantaneously, its frequency is infinite.

The bandwidth of a signal is the difference between the highest and the lowest frequencies contained in that signal.

Data & Signal



a. Bandwidth of a periodic signal



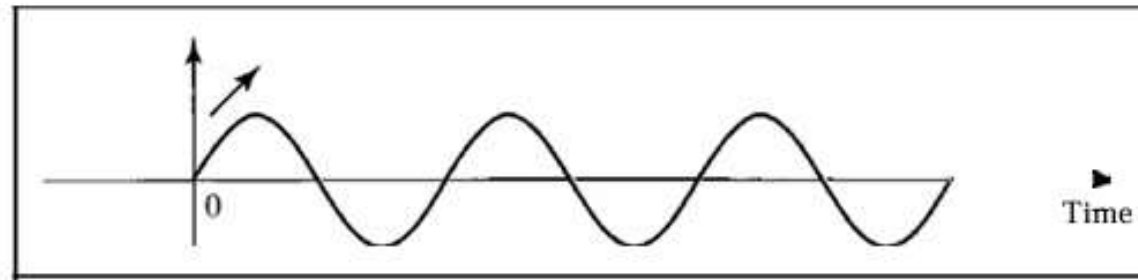
b. Bandwidth of a nonperiodic signal

Phase

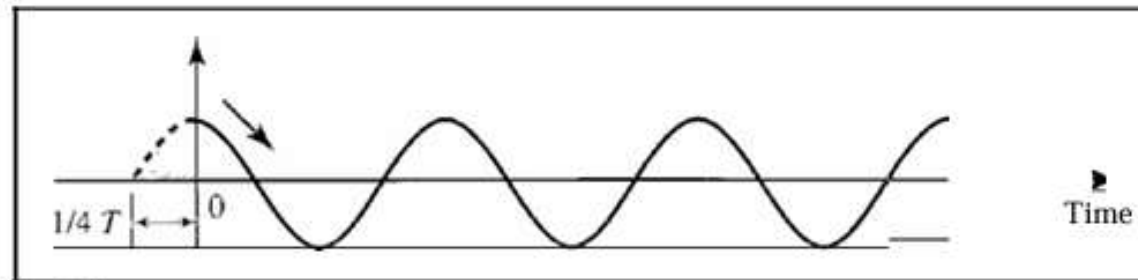
- ❖ The term phase describes the position of the waveform relative to time 0. If we think of the wave as something that can be shifted backward or forward along the time axis, phase describes the amount of that shift. It indicates the status of the first cycle

Data & Signal

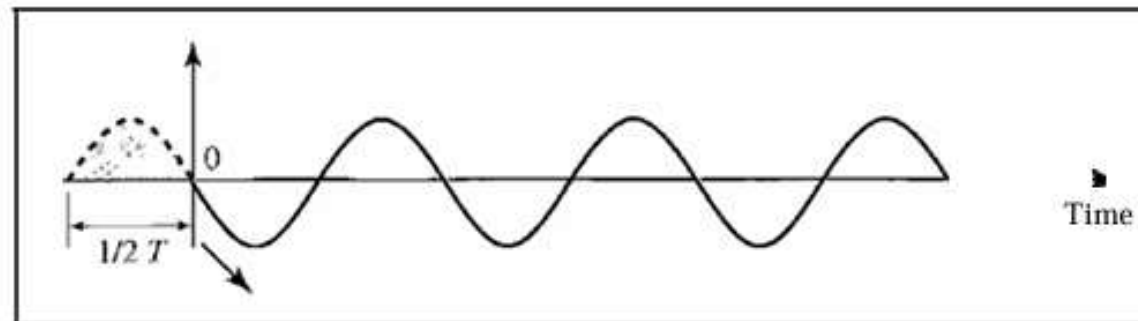
Three sine waves with the same amplitude and frequency, but different phases



a. 0 degrees



b. 90 degrees



c. 180 degrees

Wavelength

- ❖ The wavelength of a signal refers to the relationship between frequency (or period) and propagation speed of the wave through a medium.
- ❖ The wavelength is the distance a signal travels in one period.
- ❖ It is given by

$$\text{Wavelength} = \text{Propagation Speed} \times \text{Period}$$

OR

$$\text{Wavelength} = \text{Propagation Speed} \times 1/\text{Frequency}$$

- It is represented by the symbol : λ
- It is measured in meters
- It varies from one medium to another.

Digital Signal

- ❖ If n be the number of bits, then number of signal levels $L = 2^n$
- ❖ In general, if a signal has L levels, then each level need **$\text{Log}_2 L$** bits.

Example: Consider a digital Signal with four levels, how many bits are required per level?

Answer:

$$\text{Number of bits } (n) = \text{Log}_2 L = \text{Log}_2 4 = 2$$

Hence, 2 bits are required per level for a signal with four levels.

Digital Signal

BIT LENGTH or Bit Interval (T_b)

- ❖ It is the time required to send one bit.
- ❖ It is measured in seconds.

BIT RATE

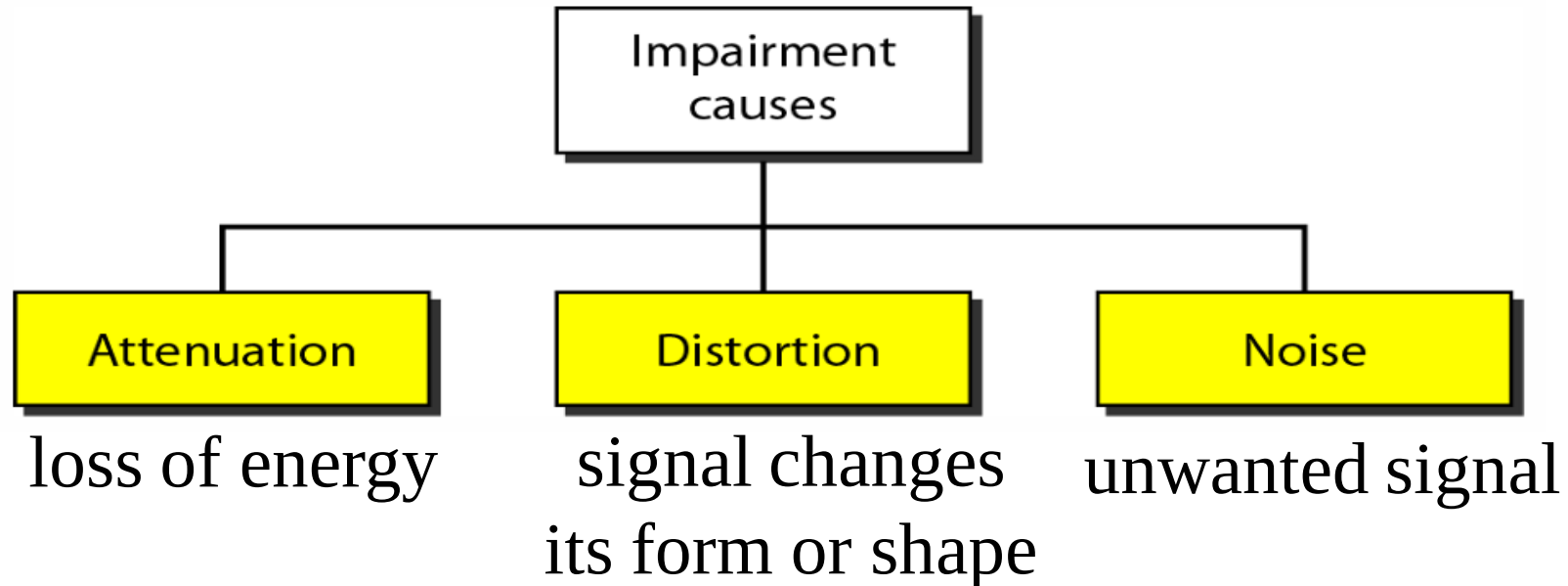
- ❖ It is the number of bits transmitted in one second.
- ❖ It is expressed as bits per second (bps).
- ❖ Relation between bit rate and bit interval can be as follows

$$\text{Bit rate} = 1 / \text{Bit interval}$$

Data & Signal

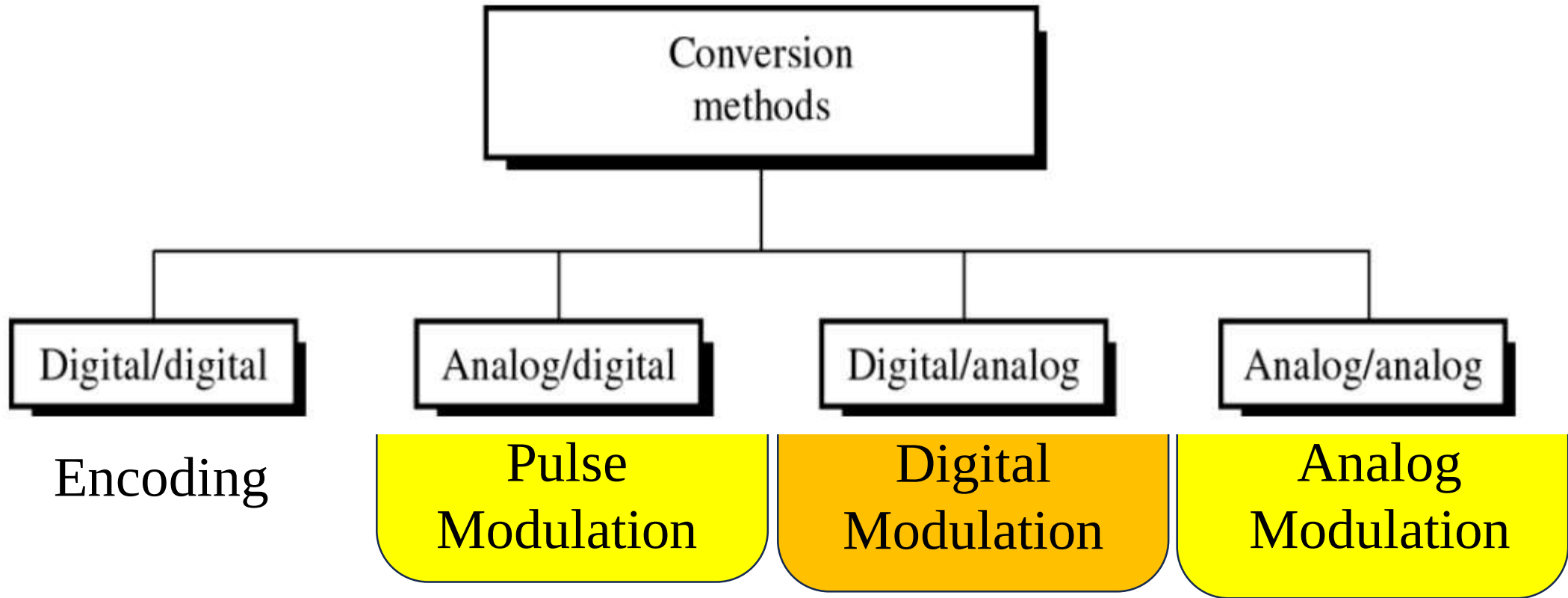
Transmission Impairment

- ❖ Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means the signal at the beginning of the medium is not same as the signal at the end of the medium.



Encoding and Modulation

- ❖ The data must be transformed into signals to send them from one place to another.

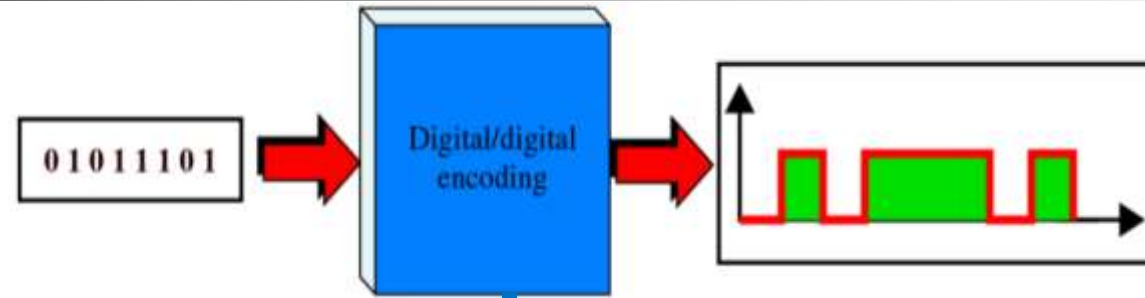


Analog Communication

Digital Communication

Encoding

**Digital data to
digital signal:
Encoding**



Line Codes

Block Codes

Scrambling

Unipolar

Polar

Bipolar

NRZ

RZ

Biphase

NRZ-L

NRZ-I

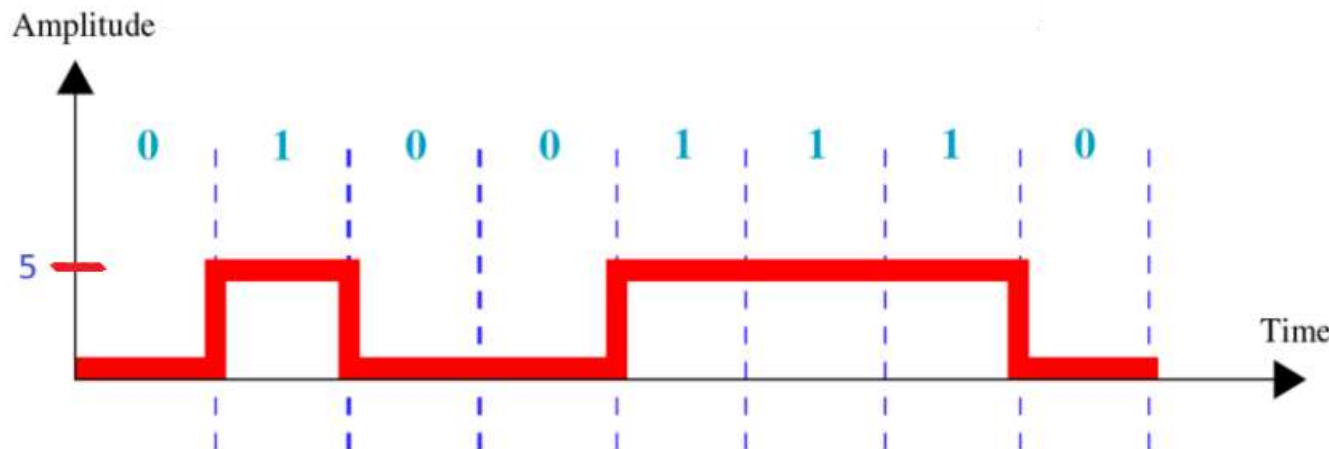
Manchester

Differential
Manchester

Encoding and Modulation

Unipolar

- ❖ Unipolar encoding is a digital encoding method that uses a single voltage level to represent binary data:
 - ✓ Binary 1 - a high voltage is transmitted, typically represented as a positive voltage.
 - ✓ Binary 0 - no voltage is transmitted; this state is often indicated by zero voltage.



Encoding and Modulation

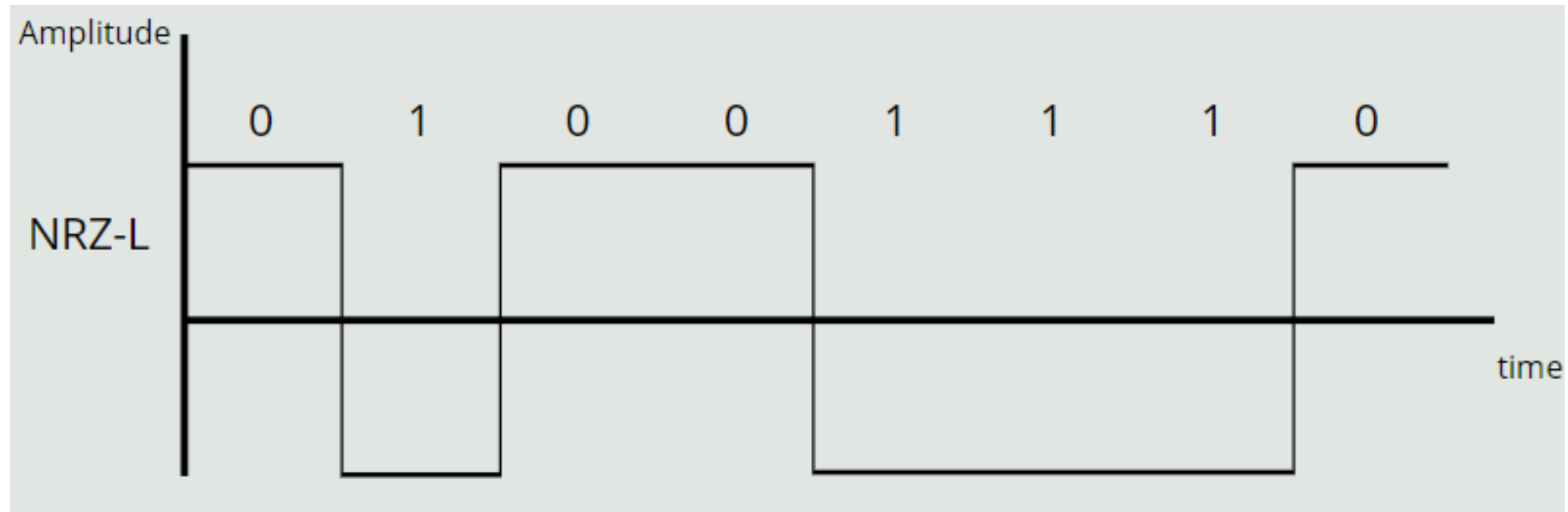
Polar Encoding

- ❖ Polar encoding uses two levels (positive and negative) of amplitude.

NRZ-L and NRZ-I Encoding

- ❖ **In NRZ-L (Non-Return to Zero – Level):**

- ✓ Binary 0 : positive voltage
- ✓ Binary 1 : negative voltage

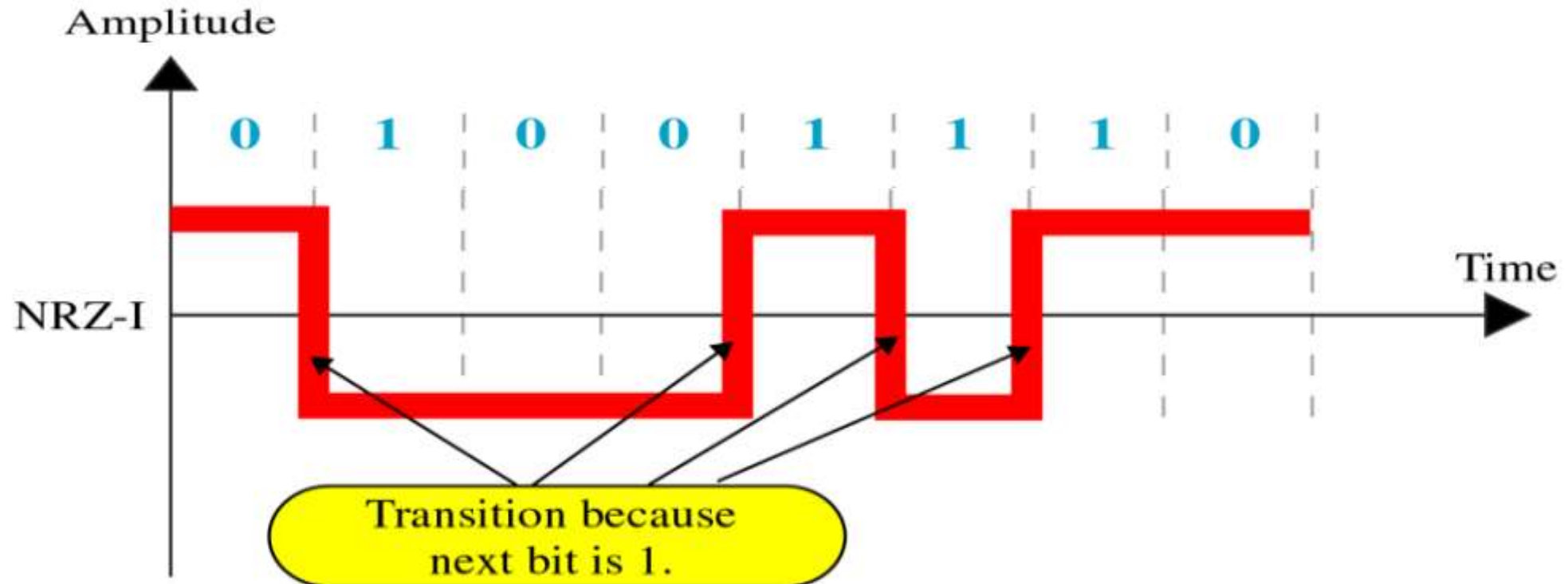


Encoding and Modulation

Polar Encoding

❖ In NRZ-I (Non-Return to Zero – Inversion):

- ✓ Binary 1: the signal is inverted
- ✓ Binary 0: no change.

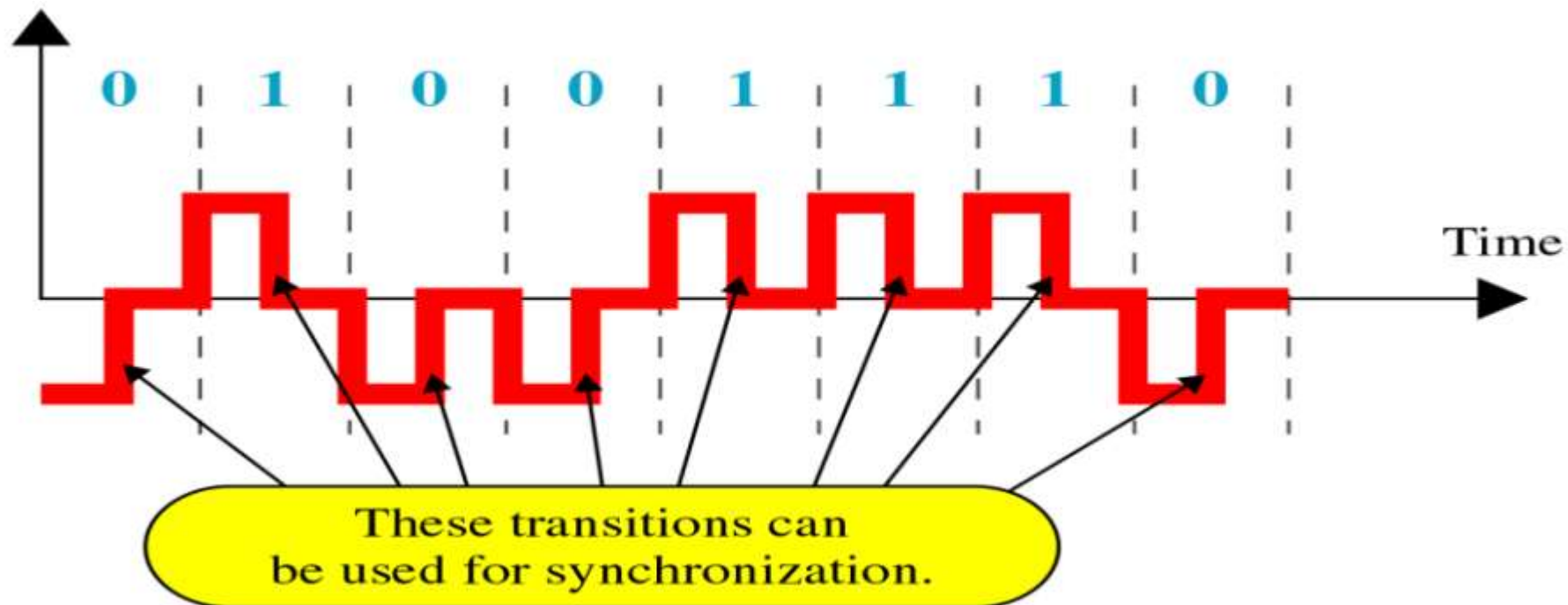


Encoding and Modulation

Polar Encoding

RZ (Return to Zero):

- ❖ RZ uses three voltage levels:
 - ✓ Binary 1 → positive and zero voltage
 - ✓ Binary 0 → negative and zero voltage
- ❖ Unlike NRZ, where only one transition occurs within a bit period, RZ encoding allows for two transitions in a single clock cycle.



Encoding and Modulation

Polar Encoding

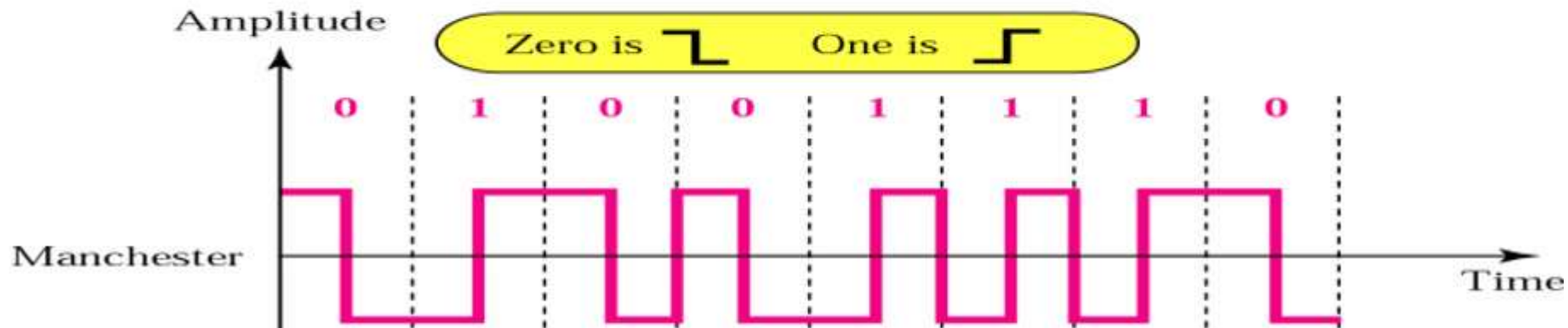
Biphase:

- ✓ Manchester and
- ✓ Differential Manchester

Manchester Encoding:

1 $\Leftrightarrow$ **low-to-high** transition

0 $\Leftrightarrow$ **high-to-low** transition



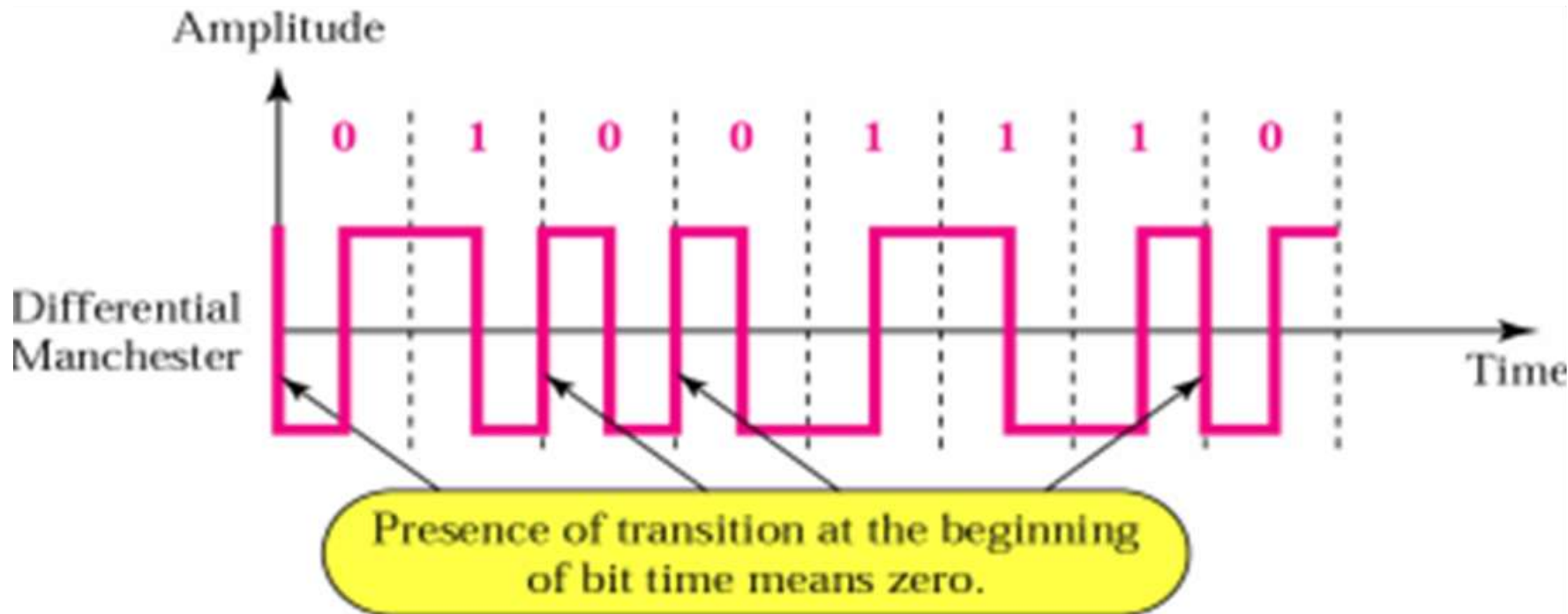
Encoding and Modulation

Polar Encoding

Differential Manchester Encoding:

- 1 $\Leftrightarrow$ **absence** of transition at the beginning of the bit interval
- 0 $\Leftrightarrow$ **presence** of transition at the beginning of the bit interval

0 – transition should happen at the beginning
starting and ending should be same



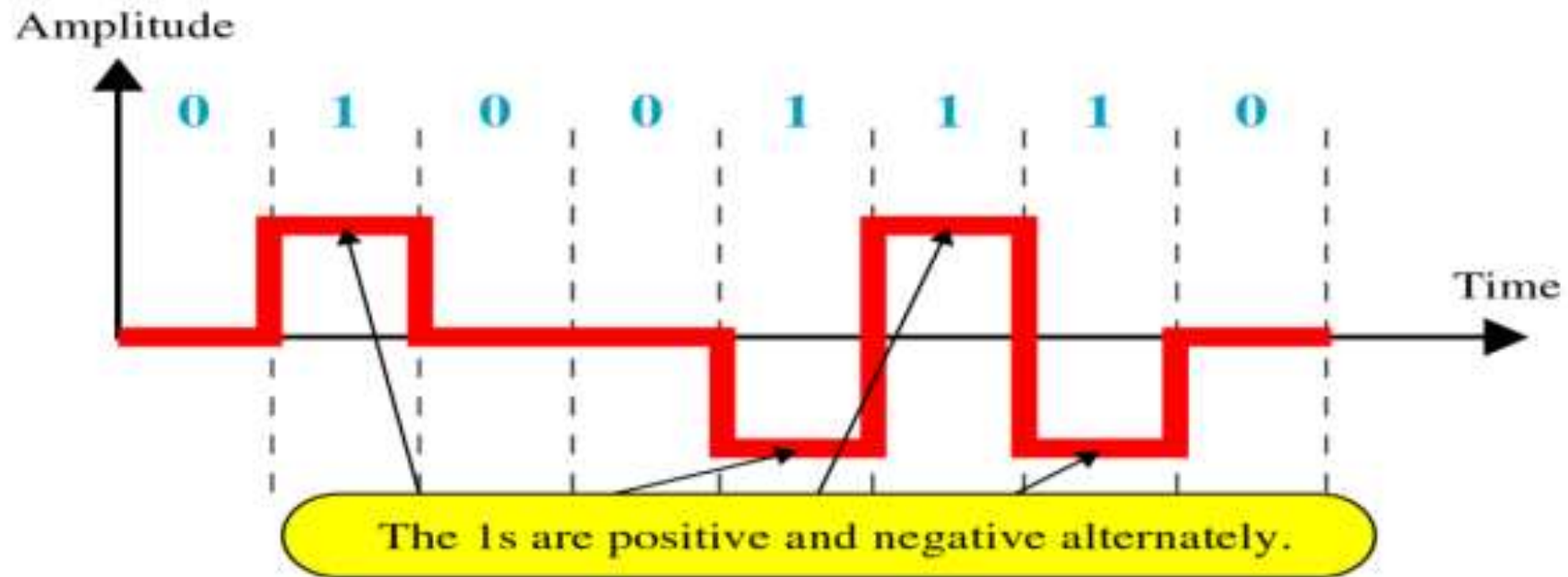
0:-[High,Low]

1:-[Low,High]

Encoding and Modulation

Bipolar Encoding

Three levels: positive, zero and negative





Encoding and Modulation

Modulation

- ❖ Modulation is the process of placing or superimposing the low frequency message over the high frequency carrier signal make it suitable for transmission over long distance.

(or)

- ❖ Changing the carrier signal with respect to the message signal

Modulation is the process of changing any one of the characteristics (Amplitude or frequency or phase angle) of (high frequency) carrier signal in accordance with the instantaneous values (amplitude) of the message (information) signal.



Encoding and Modulation

Need for Modulation

- ❖ The primary purpose of modulation in a communication system is to generate a modulated signal which is well suited to the characteristics of transmission medium.
- ❖ The audio frequency message signal carrying information can't be transmitted for distance directly without modulation.
- ❖ According to inverse square law, the fading (reduction in signal strength) of the signal directly proportional to the square of the distance it travels. So, the signal strength decreases naturally.



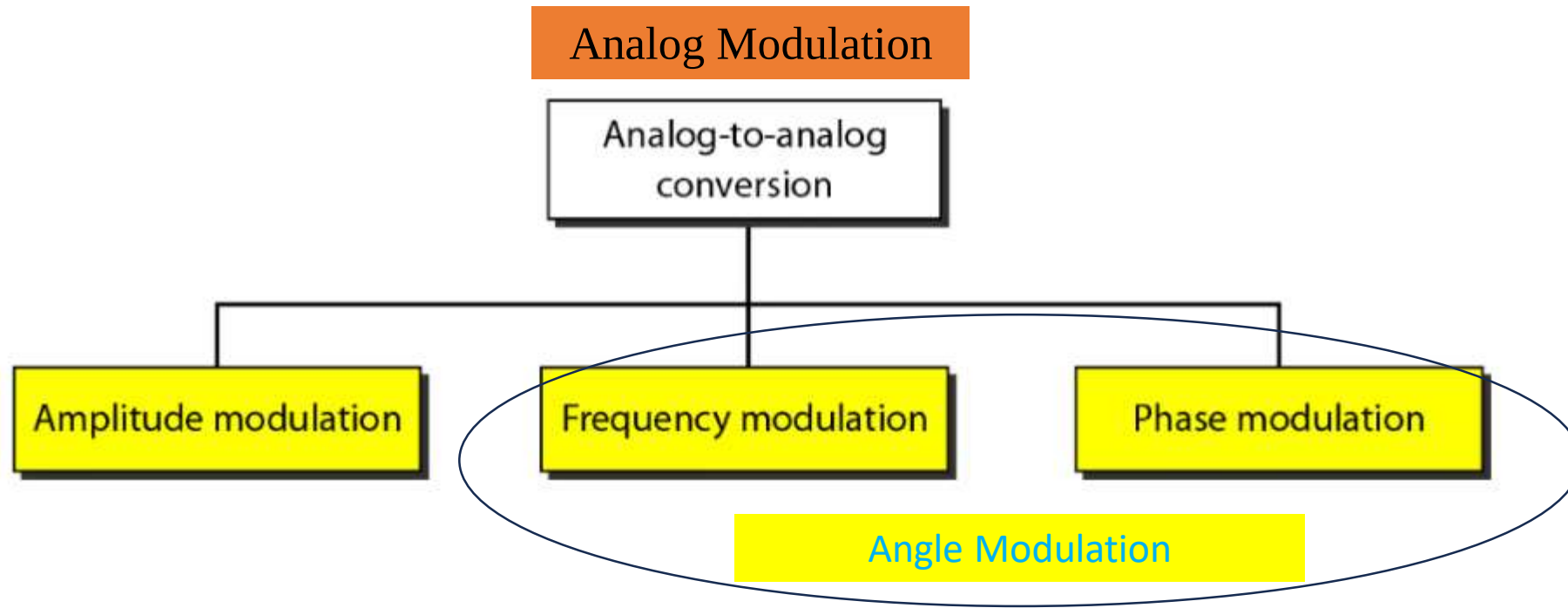
Encoding and Modulation

Need for Modulation

- ❖ To reduce the antenna height (Height is inversely proportional to the frequency)
- ❖ To reduce the interference, noise & distortions made when we transmit the signals with nearly same frequency in the audio frequency range (20-20k) Hz.
- ❖ To multiplex the more number of signals
- ❖ To reduce the complexity of the transmission system
- ❖ To increase the bandwidth of the signal

Encoding and Modulation

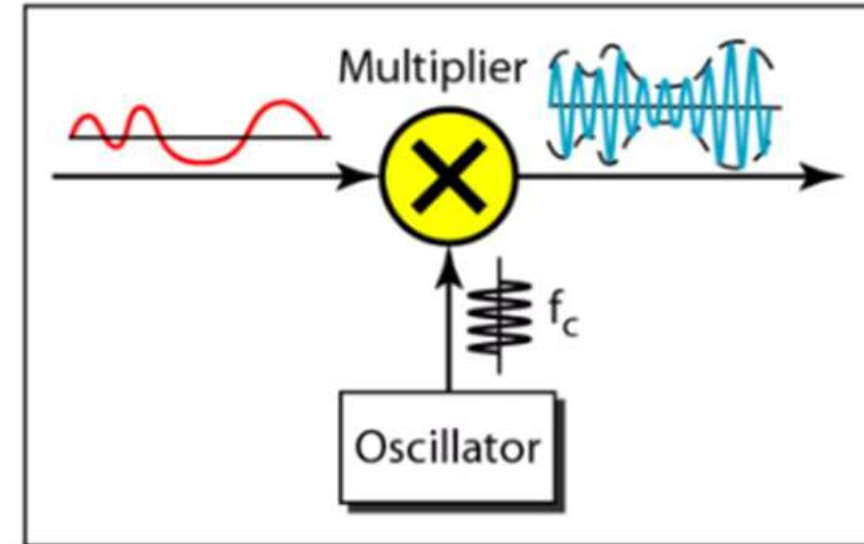
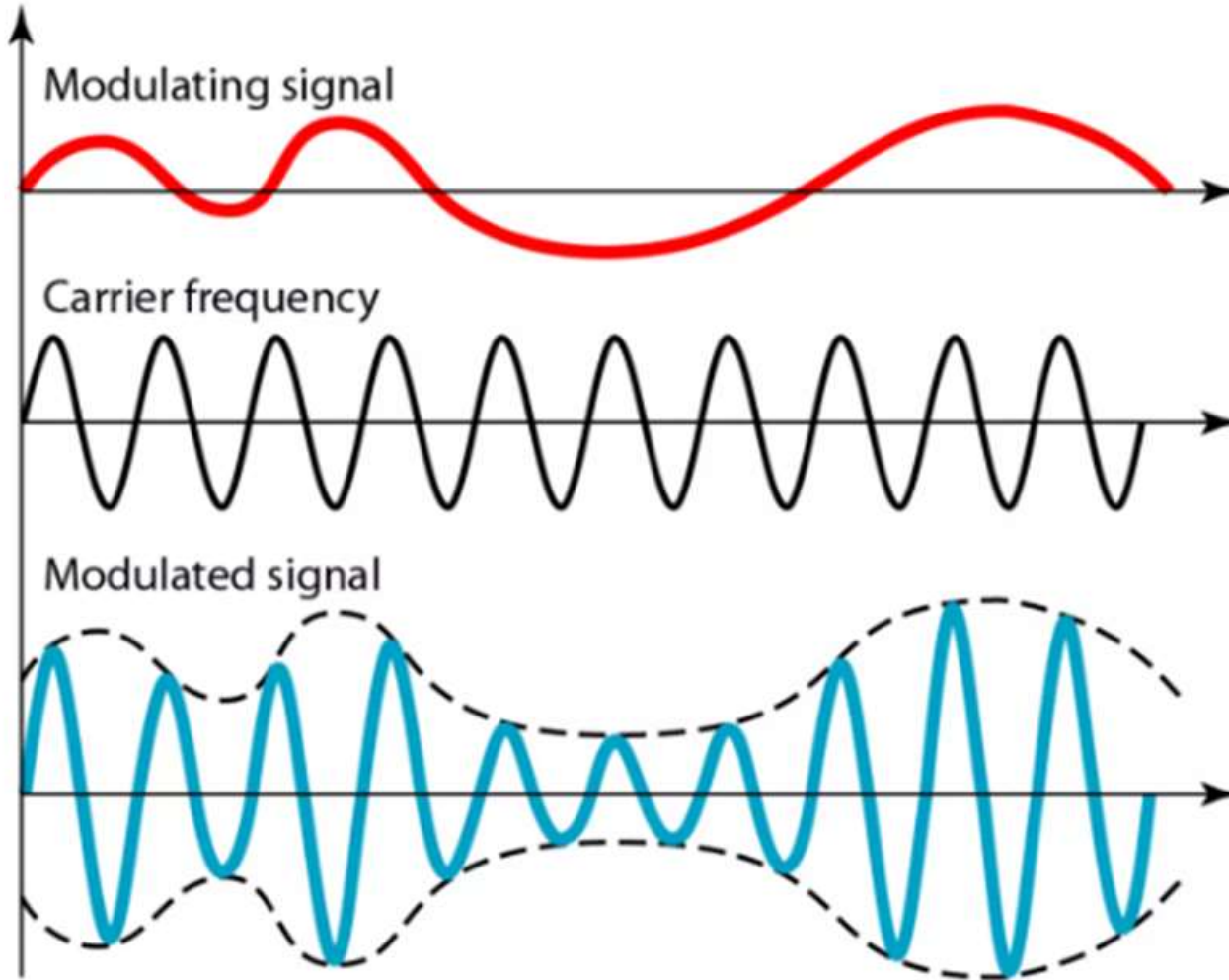
Types of Analog Modulation



The carrier signal amplitude, frequency or phase is changed in accordance with the modulating signal to obtain AM, FM or PM respectively.

Encoding and Modulation

Amplitude Modulation

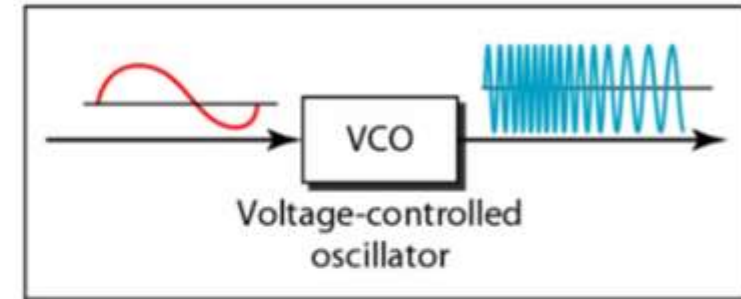
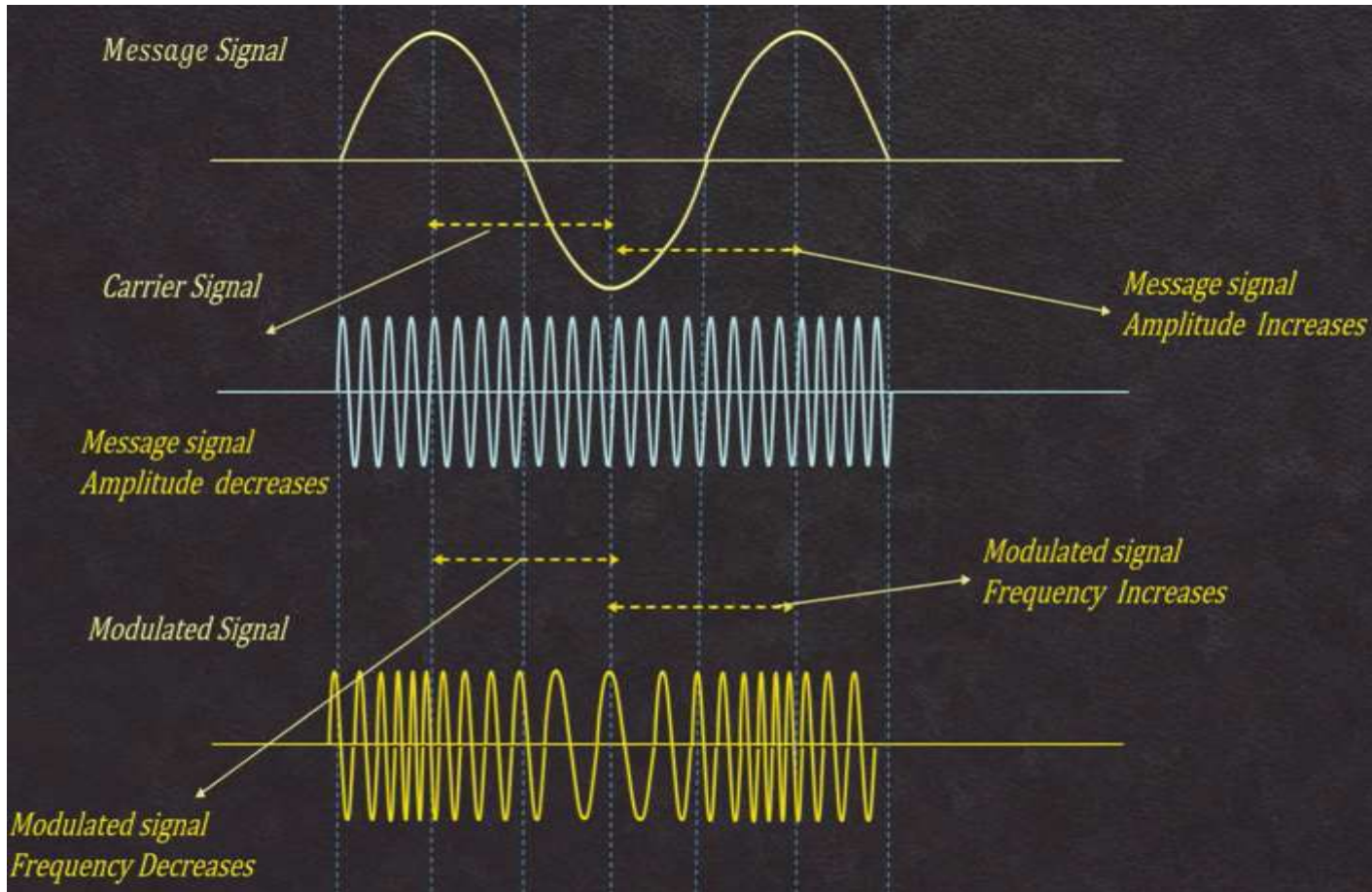


The total bandwidth required for AM can be determined from the bandwidth of the audio signal: $B_{AM} = 2B$.

Encoding and Modulation

Frequency Modulation

- ❖ The modulating signal changes the frequency of the carrier signal

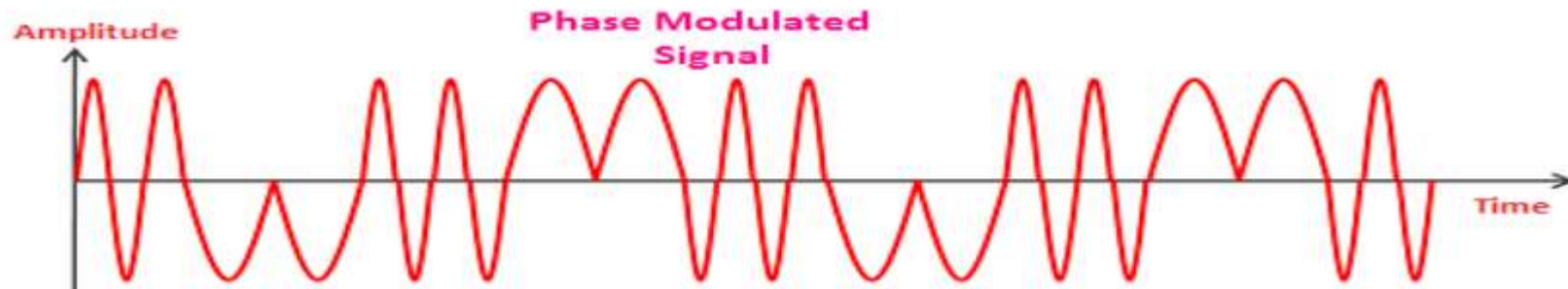
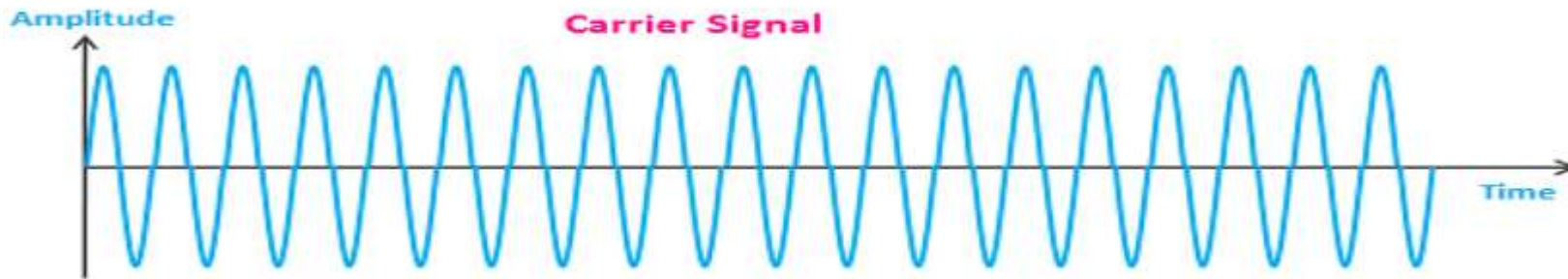
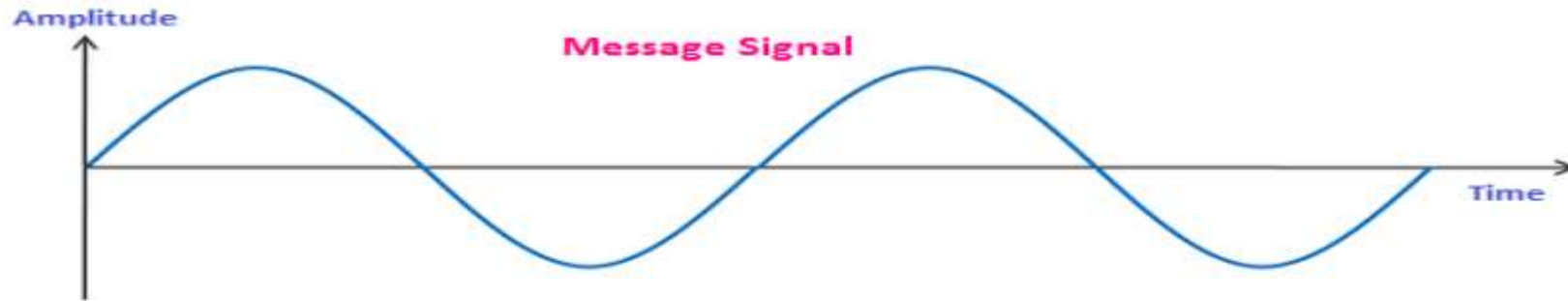


- The bandwidth for FM is high
- It is approx. 10x the signal frequency

Encoding and Modulation

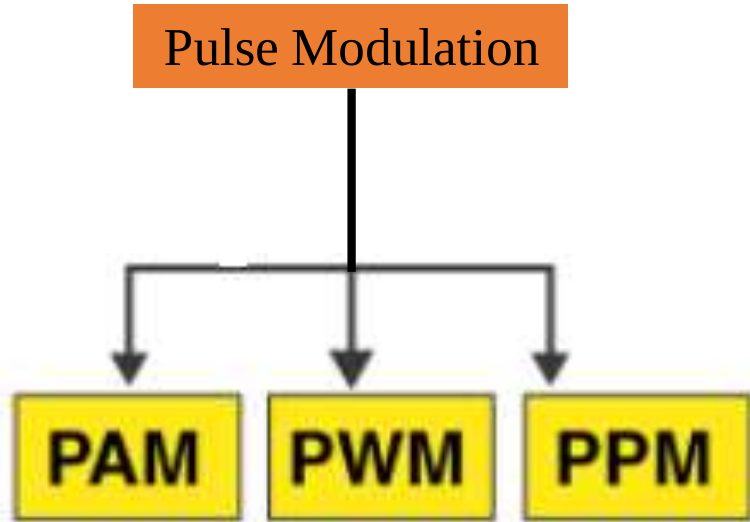
Phase Modulation

- ❖ The modulating signal changes the phase of the carrier signal



Encoding and Modulation

Types of Pulse Modulation



PAM – Pulse Amplitude Modulation

PWM – Pulse Width Modulation

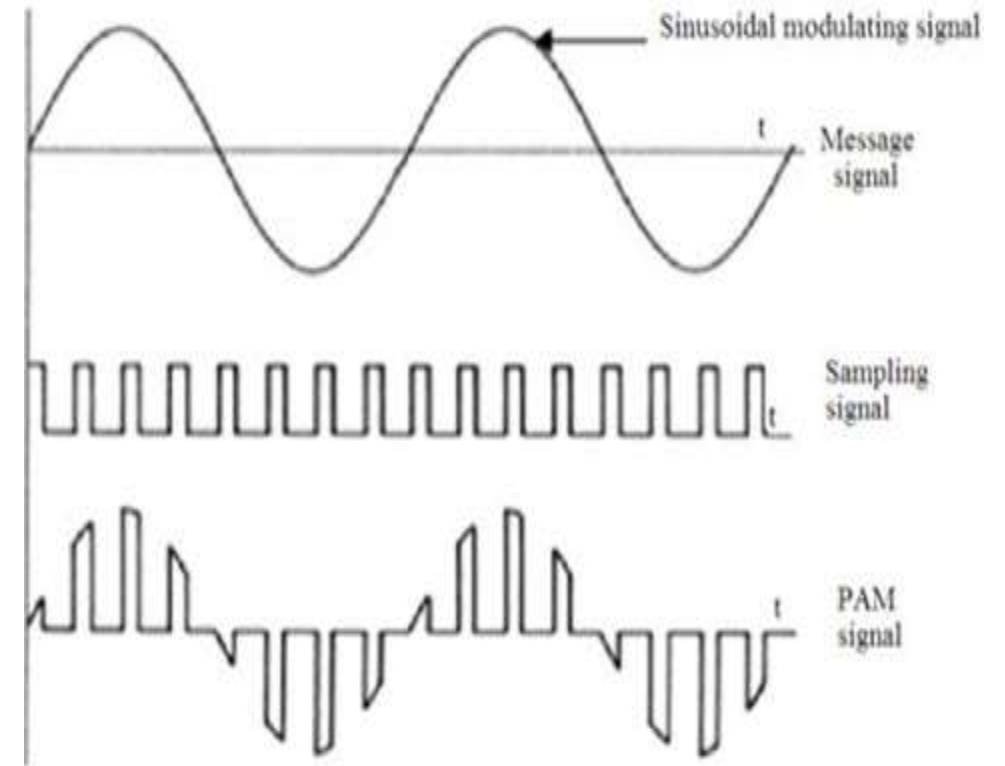
PPM – Pulse Position Modulation

The amplitude, duration and position of the pulse is varied in accordance with the instantaneous value of the baseband modulating signal or message signal

Encoding and Modulation

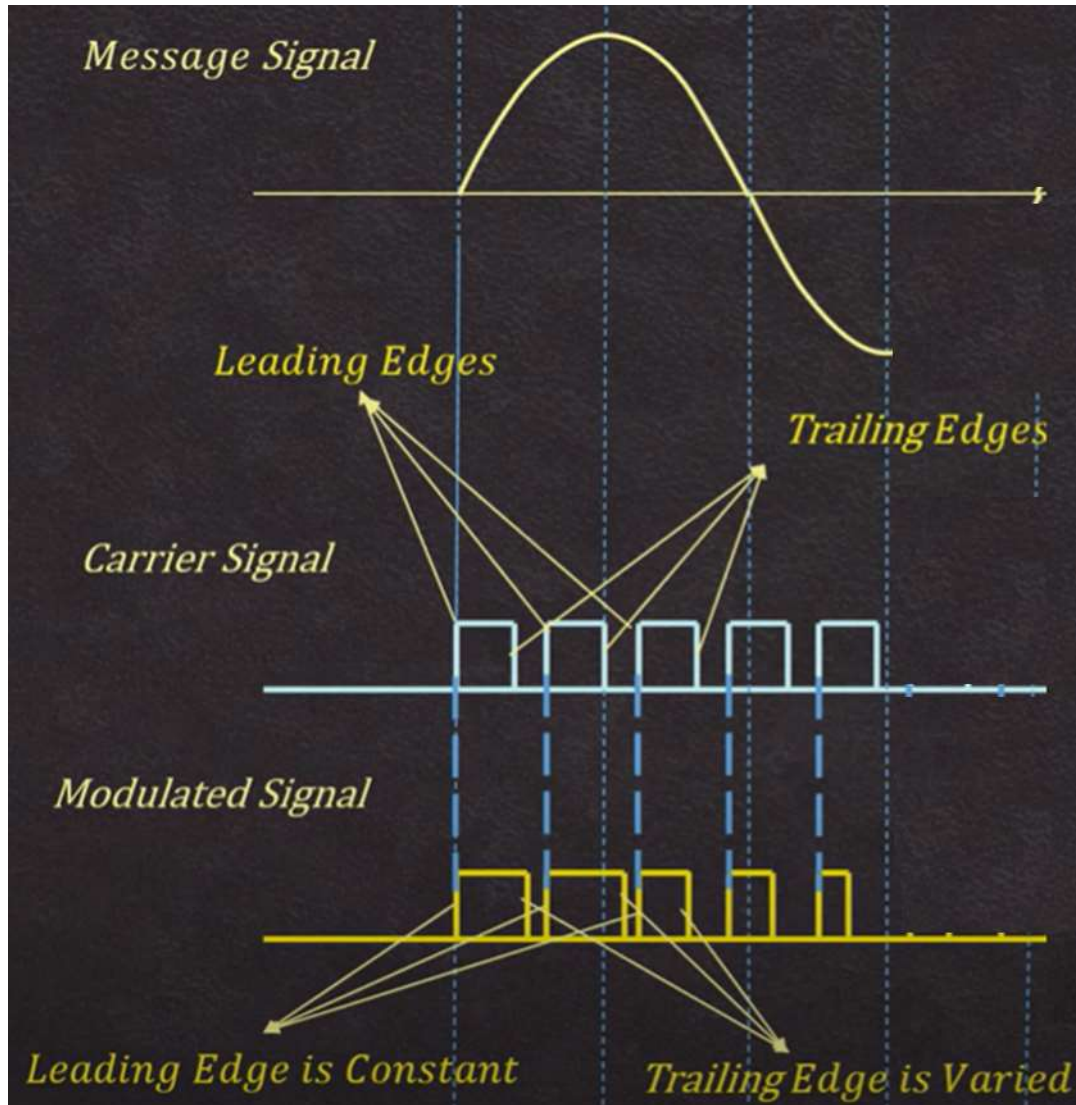
Pulse Amplitude Modulation

- ❖ The amplitude of each pulse is directly proportional to modulating signal amplitude at the time of pulse occurrence
- ❖ Then follows the amplitude of the pulse for the rest of the half cycle



Encoding and Modulation

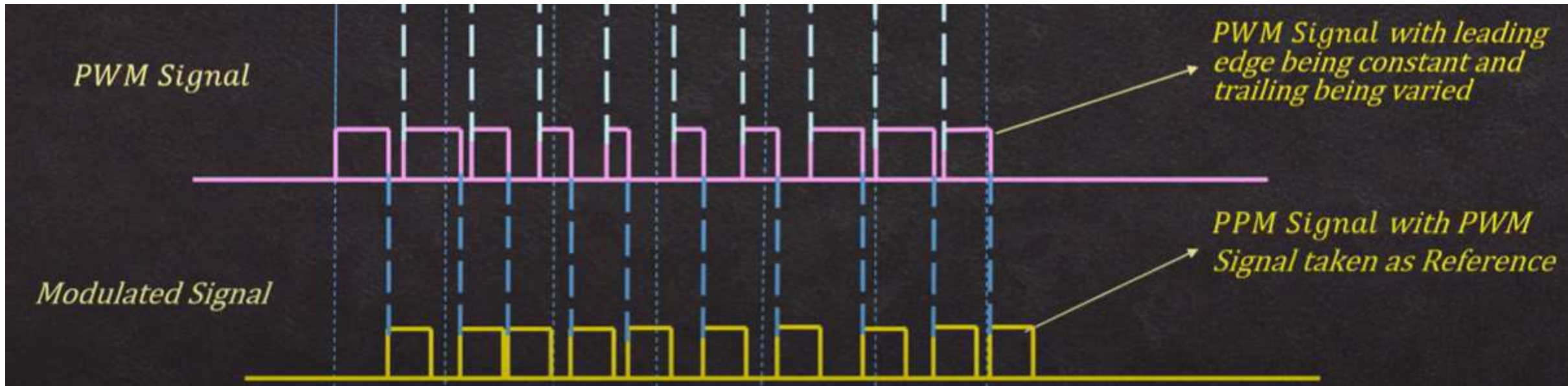
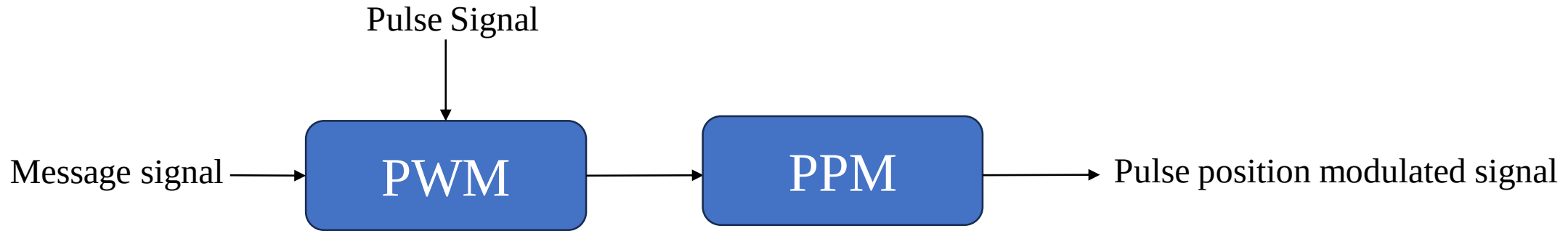
Pulse Width Modulation



- ❖ The width or duration of the pulse is varied in accordance with the instantaneous value of the modulating signal
- ❖ The leading edge of the pulse being constant, the trailing edge varies according to the message signal.

Encoding and Modulation

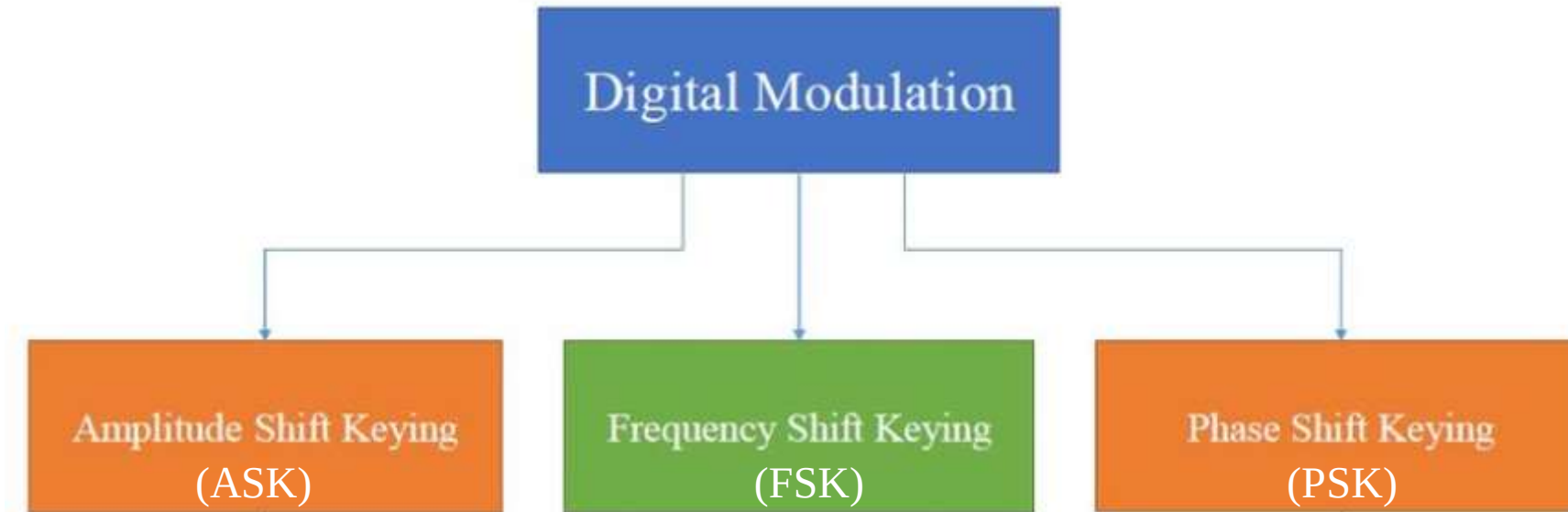
Pulse Position Modulation



- ❖ The position of the pulse is varied in accordance with the instantaneous value of the modulating signal

Encoding and Modulation

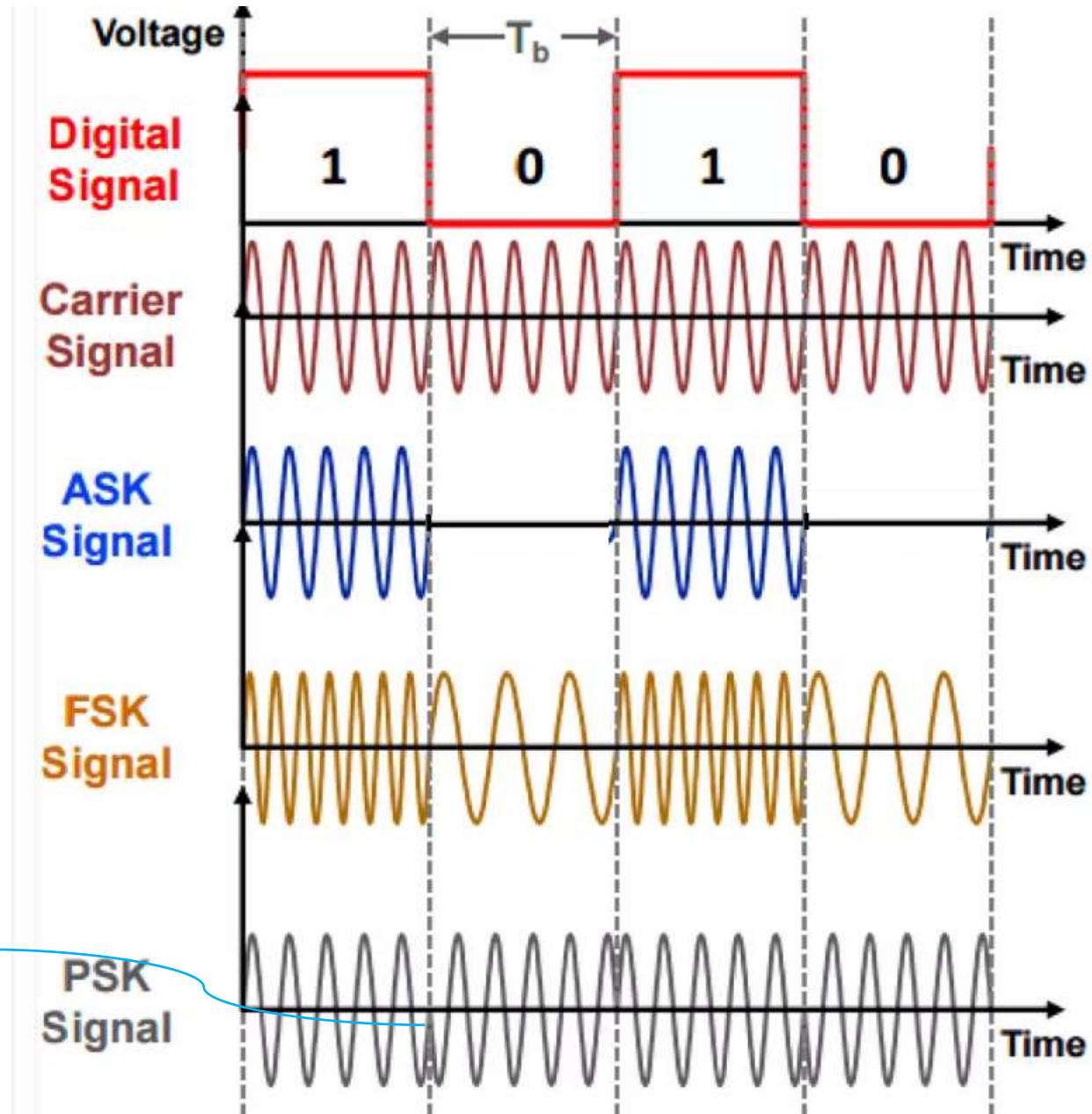
Types of Digital Modulation



Digital modulation is the process of encoding a digital information into the analog signal with varying amplitude, phase, or frequency of the high frequency carrier signal.

Encoding and Modulation

- ❖ In ASK, the amplitude of the signal is changed in response to the digital information. Binary 1 – the amplitude of the carrier signal remains same and Binary 0 – the amplitude is zero
- ❖ In FSK, the frequency of the signal is changed in response to the digital information. Binary 1 – the frequency of the carrier signal is decreased and Binary 0 – the frequency of the carrier signal is increased
- ❖ Binary 0 – phase of the carrier signal is shifted by 180°





Data Rate Limits

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

- 1. The bandwidth available*
- 2. The level of the signals we use*
- 3. The quality of the channel (the level of noise)*



Nyquist Bit Rate

The Nyquist bit rate formula defines the theoretical maximum bit rate for a noiseless channel

$$\text{Bit rate} = 2 \times \text{Bandwidth} \times \log_2 L$$

Where,

Bitrate is the bitrate of the channel in bits per second

Bandwidth is the bandwidth of the channel

L is the number of signal levels.



Nyquist Bit Rate

Example:

What is the maximum bit rate of a noiseless channel with a bandwidth of 5000 Hz transmitting a signal with two signal levels.

Solution:

The bit rate for a noiseless channel according to Nyquist Bit rate can be calculated as follows:

$$\begin{aligned}\text{Bit Rate} &= 2 \times \text{Bandwidth} \times \log_2 L \\ &= 2 \times 5000 \times \log_2 2 \\ &= 10000 \text{ bps}\end{aligned}$$

$$\log_2 8 = \log_2 2^3 = 3 \log_2 2 = 3$$



Shannon's Capacity Theorem

The Shannon Capacity defines the theoretical maximum bit rate for a noisy channel

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

Where,

Capacity is the capacity of the channel in bits per second

Bandwidth is the bandwidth of the channel

SNR is the Signal to Noise Ratio

Shannon Capacity for calculating the maximum bit rate for a noisy channel does not consider the number of levels of the signals being transmitted as done in the Nyquist bit rate.



Shannon's Capacity Theorem

Example:

Calculate the bit rate for a noisy channel with SNR 300 and bandwidth of 3000Hz

Solution:

The bit rate for a noisy channel according to Shannon Capacity can be calculated as follows:

$$\begin{aligned}\text{Capacity} &= \text{bandwidth} \times \log_2 (1 + \text{SNR}) \\ &= 3000 \times \log_2 (1 + 300) \\ &= 3000 \times \log_2 (301) \\ &= 3000 \times 8.23 = 24,690 \text{ bps} = 24.69 \text{ kbps}\end{aligned}$$



Switching techniques- TDM, FDM

- ❖ **Multiplexing** is the process of combining multiple signals into one signal, over a shared medium.
- ❖ The process of multiplexing divides a communication channel into several number of logical channels, allotting each one for a different message signal or a data stream to be transferred. The device that does multiplexing can be called as **Multiplexer or MUX**.
- ❖ The reverse process, i.e., extracting the number of channels from one, which is done at the receiver is called as de-multiplexing. The device that does de-multiplexing can be called as **de-multiplexer or DEMUX**.
- ❖ **Classification:**
 - ✓ Time Division Multiplexing (TDM)
 - ✓ Frequency Division Multiplexing (FDM)



Switching techniques- TDM, FDM

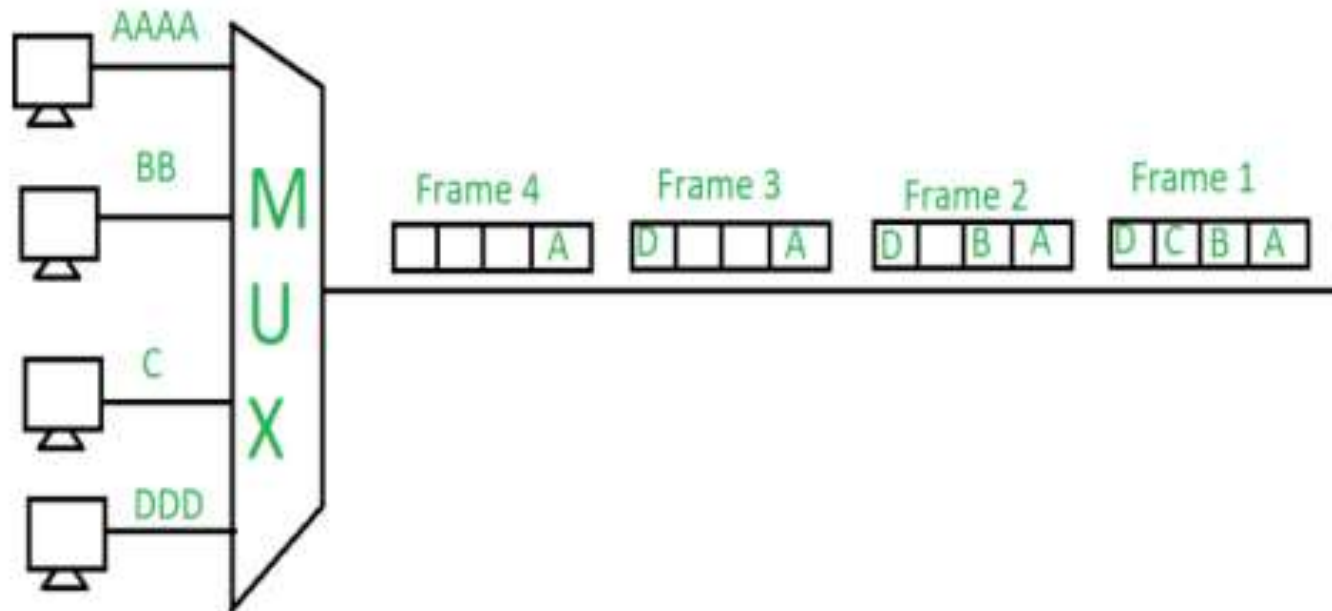
Time Division Multiplexing (TDM)

- ❖ TDM is a multiplexing technique in which each signal is assigned a different time slot for transmission.
- ❖ In TDM users are allowed to utilize the total available bandwidth on time sharing basis.
- ❖ Here the time domain is divided into several recurrent slots of fixed length, and each signal is allotted a time slot on a round-robin basis.
- ❖ Types:
 - ✓ Synchronous TDM
 - ✓ Asynchronous TDM

Switching techniques- TDM, FDM

Synchronous TDM

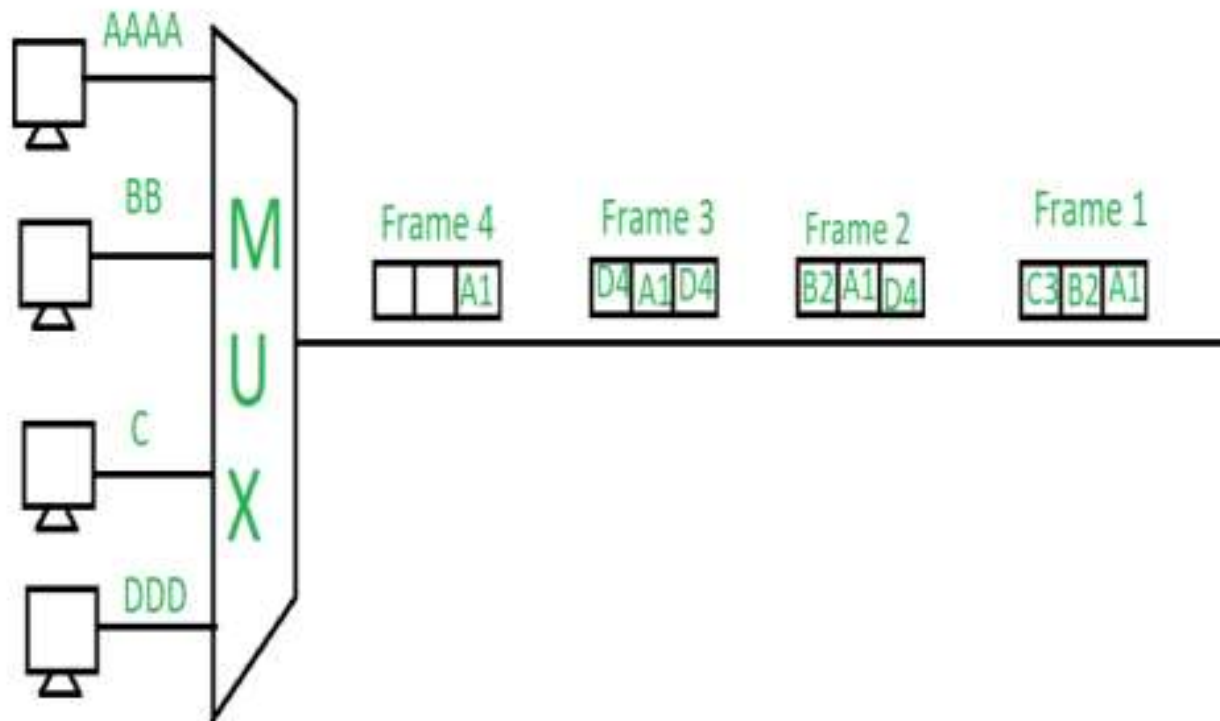
- ❖ The time slots are pre-assigned and fixed.
- ❖ This slot is even given if the source is not ready with data at this time. In this case the slot is transmitted empty.
- ❖ It is used for multiplexing digitized voice stream.



Switching techniques- TDM, FDM

Asynchronous TDM

- ❖ The slots are allocated dynamically depending on the speed of source or their ready state.
- ❖ It dynamically allocates the time slots according to different input channel's needs, thus saving the channel capacity.





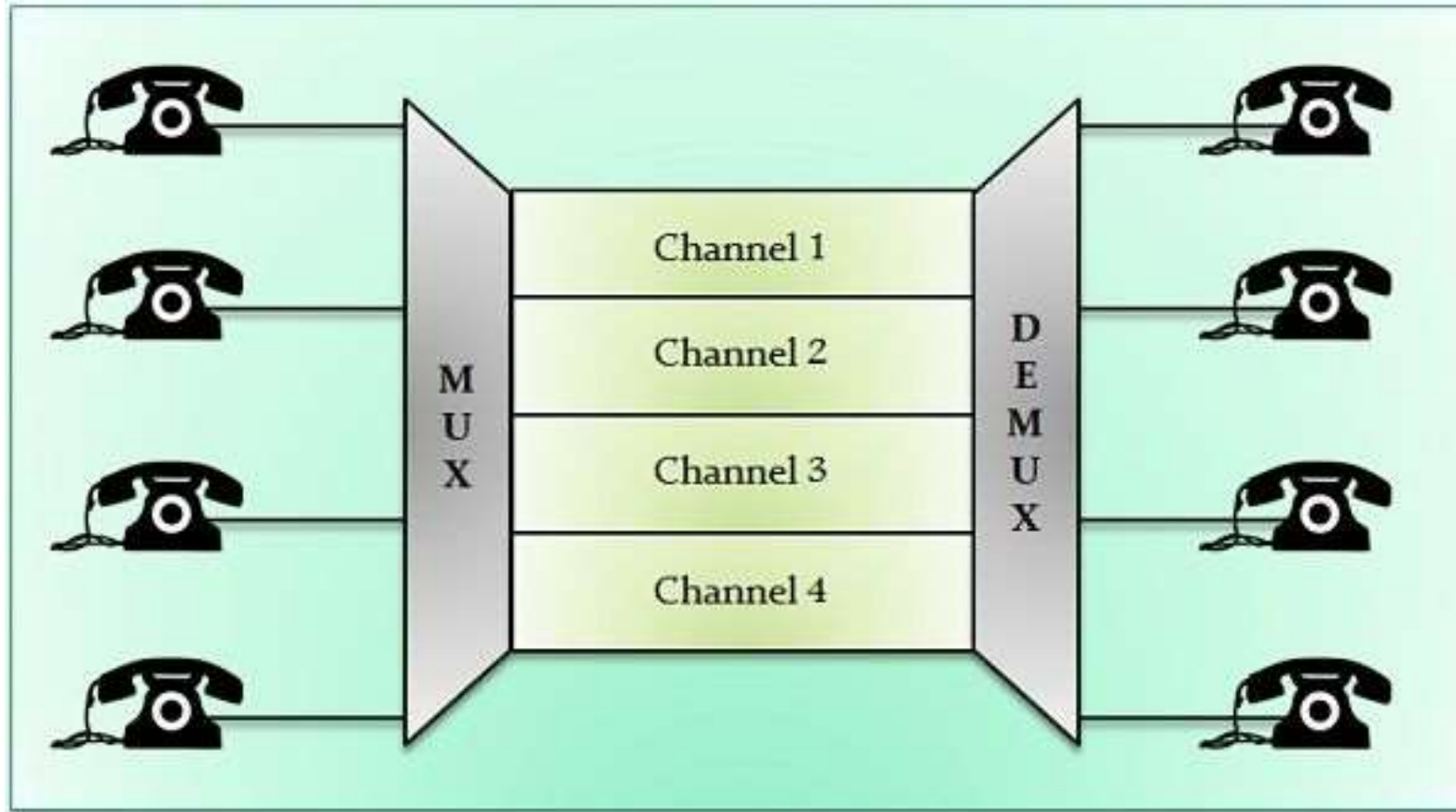
Switching techniques- TDM, FDM

FDM

- ❖ In this a number of signals are transmitted at the same time, and each source transfers its signals in the allotted frequency range.
- ❖ There is a suitable frequency gap (Guard Band) between the 2 adjacent signals to avoid over-lapping/collision.
- ❖ The frequency spectrum is divided into several logical channels, in which every user feels that they possess a particular bandwidth. A number of signals are sent simultaneously on the same time allocating separate frequency band or channel to each signal.
- ❖ It is used in radio and TV transmission.

Switching techniques- TDM, FDM

FDM





Switching techniques- TDM, FDM

Comparison between TDM and FDM

BASIS FOR COMPARISON	TDM	FDM
Basic	Times scale is shared.	Frequency is shared.
Used with	Digital signals and analog signals	Analog signals
Necessary requirement	Sync Pulse	Guard Band
Interference	Low or negligible	High
Circuitry	Simpler	Complex
Utilization	Efficiently used	Ineffective